

$$\textcircled{1} a) \frac{\partial F_1}{\partial y} = 2 \sin(\alpha x) \quad , \quad \frac{\partial F_2}{\partial x} = \alpha \sin(\alpha x)$$

$$F \text{ potenciál na } \mathbb{R}^2 \Leftrightarrow \frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x} \text{ na } \mathbb{R}^2$$

$$\Leftrightarrow (2-\alpha) \sin(\alpha x) = 0 \quad \forall x \in \mathbb{R} \Leftrightarrow \alpha \in \{0, 2\}.$$

b)

- $\alpha = 0$... $F(x, y) = (0, -1)$

Potenciál F je v tomto případě $f(x, y) = -y + k$.

$\& f(0, 2) = 0$ plyne $k = 2$. Tedy

$$f(x, y) = -y + 2$$

- $\alpha = 2$... $F(x, y) = (2y \sin(2x), 2y - \cos(2x))$

$$\frac{\partial f}{\partial x} = 2y \sin(2x) \quad (1)$$

$$\frac{\partial f}{\partial y} = 2y - \cos(2x) \quad (2)$$

$\& (1)$ máme $f(x, y) = -y \cos(2x) + C(y)$. Dosazením do (2) obdržíme $C'(y) = 2y$. Tedy $C(y) = y^2 + k$.

Protože $f(0, 2) = 0$, je $0 = -2 + 4 + k$. Odtud

$$k = -2. \text{ Proto } f(x, y) = -y \cos(2x) + y^2 - 2.$$

$$\textcircled{2} \frac{\partial f}{\partial x} = 2x - 2 = 0 \Leftrightarrow \underline{x = 1}.$$

$$\frac{\partial f}{\partial y} = y^3 - 2y(2z-1) = 0 \Leftrightarrow y(y^2 - 4z + 2) = 0$$

$$\Leftrightarrow y = 0 \text{ me } y^2 = 4z - 2.$$

$$\frac{\partial f}{\partial z} = 4z - 2y^2 + 1 = 0 \quad (*)$$

- $y = 0$: $\mathcal{L} (*)$ me $z = -\frac{1}{4}$.

- $y^2 = 4z - 2$: $\mathcal{L} (*)$ me $4z - 2(4z - 2) + 1 = 0$.

~~Proble~~ $-4z + 5 = 0 \Leftrightarrow \underline{z = \frac{5}{4}}$

Tedy $y^2 = 5 - 2 = 3 \Leftrightarrow \underline{y = \pm\sqrt{3}}$

$$H_f(x, y, z) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3y^2 - 2(2z-1) & -4y \\ 0 & -4y & 4 \end{pmatrix}$$

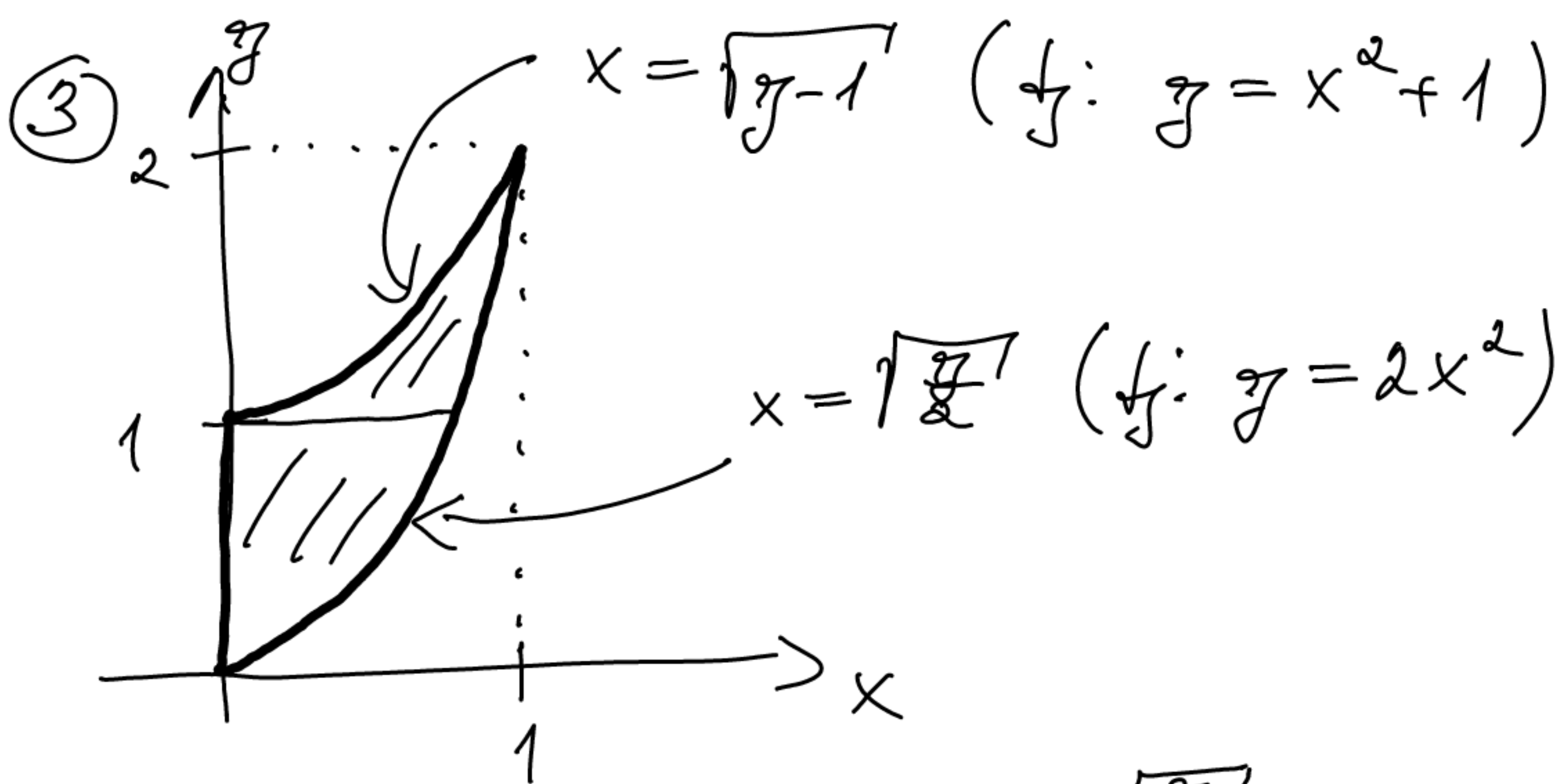
- $H_f(1, 0, -\frac{1}{4}) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$ je pozit. def $\Rightarrow (1, 0, -\frac{1}{4})$ je lokal. min.

- $H_f(1, \pm\sqrt{3}, \frac{5}{4}) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 6 & \mp 4\sqrt{3} \\ 0 & \mp 4\sqrt{3} & 4 \end{pmatrix} \dots 2 > 0, \begin{vmatrix} 2 & 0 \\ 0 & 6 \end{vmatrix} > 0,$

$$\det H_f(1, \pm\sqrt{3}, \frac{5}{4}) = 2 \begin{vmatrix} 6 & \mp 4\sqrt{3} \\ \mp 4\sqrt{3} & 4 \end{vmatrix} = 2(24 - 48) = -48 < 0$$

$$\Rightarrow H_f(1, \pm\sqrt{3}, \frac{5}{4}) \text{ je indefinitus} \Rightarrow (1, \pm\sqrt{3}, \frac{5}{4})$$

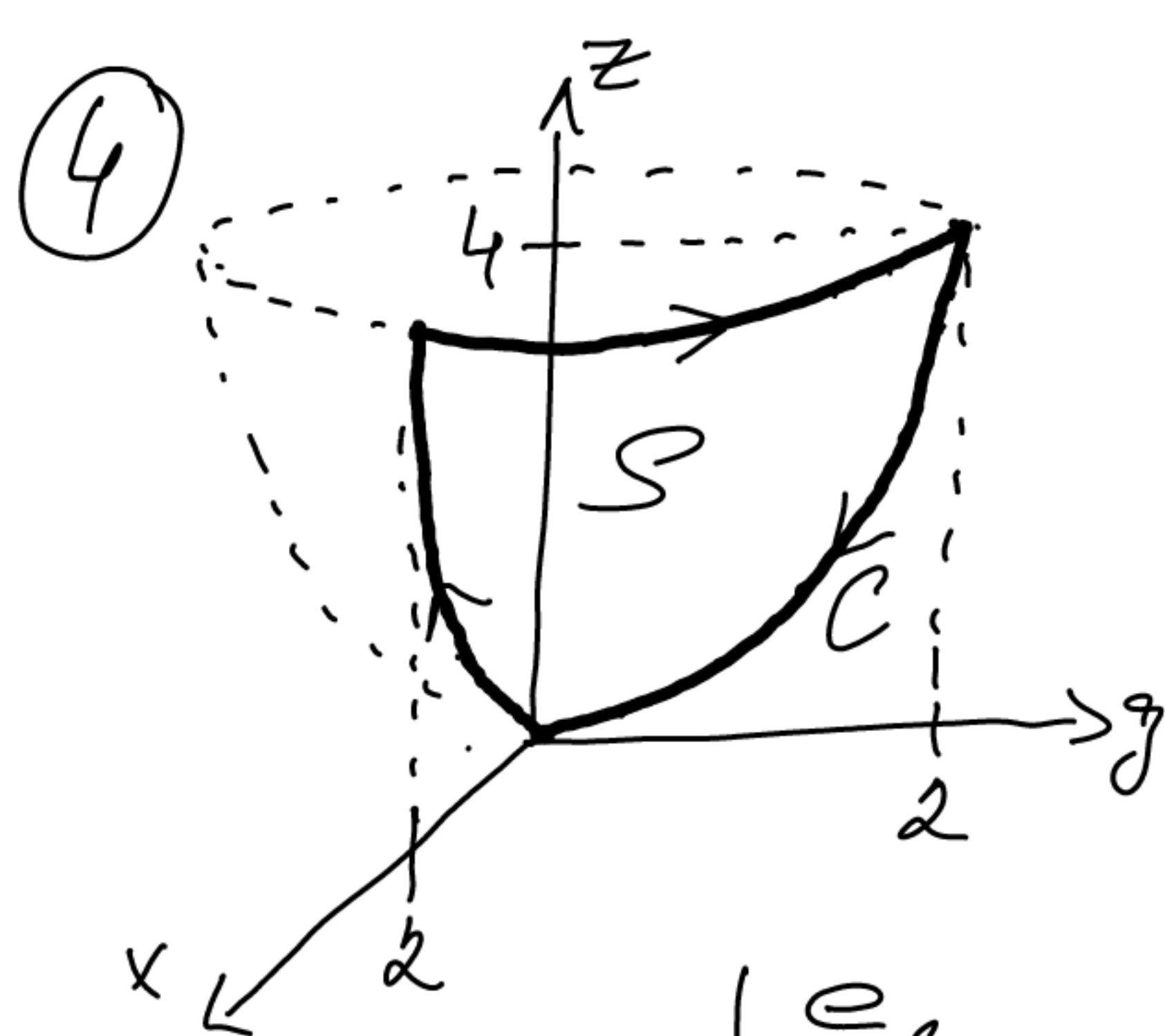
jeon reallove body.



$$\int_0^1 \int_0^{\sqrt{\frac{y}{2}}} e^{-x^3+3x} dx dy + \int_1^2 \int_{\sqrt{y-1}}^{\sqrt{\frac{y}{2}}} e^{-x^3+3x} dx dy =$$

$$= \int_0^1 \int_{2x^2}^{x^2+1} e^{-x^3+3x} dy dx = \int_0^1 (-x^2+1) e^{-x^3+3x} dx$$

$$= \left[\frac{e^{-x^3+3x}}{3} \right]_0^1 = \frac{e^2-1}{3}$$



Param. plochy S :

$$\Phi(u, v) = (u \cos v, u \sin v, u^2),$$

$$u \in [0, 2], v \in [0, \frac{\pi}{2}].$$

$$\frac{\partial \Phi}{\partial u} \times \frac{\partial \Phi}{\partial v} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ \cos v & \sin v & 2u \\ -u \sin v & u \cos v & 0 \end{vmatrix} =$$

$$= (-2u^2 \cos v, -2u^2 \sin v, u)$$

Param. Φ indukuje souhlasnou orientaci plochy S a orientaci křivky C .

$$\nabla \times F = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} = \overbrace{(2z - 2z, -y, z)}^{=0} = (0, -y, z)$$

$$\Rightarrow \int_C F = \int_S \nabla \times F = \int_0^{\frac{\pi}{2}} \int_0^2 \underbrace{(0, -u \sin v, u^2) \cdot (-2u^2 \cos v, -2u^2 \sin v, u)}_{= 2u^3 \sin^2 v + u^3} du dv$$

$$= \int_0^{\frac{\pi}{2}} \left[\frac{u^4}{4} \right]_0^2 (1 + 2 \sin^2 v) dv = 4 \int_0^{\frac{\pi}{2}} 2 - \cos(2v) dv$$

$$= 4\pi - 4 \left[\frac{\sin(2v)}{2} \right]_0^{\frac{\pi}{2}} = 4\pi.$$

$$\begin{aligned}
 \textcircled{5} \quad a_\ell &= \frac{1}{2} \int_{-2}^2 f(t) \cos\left(\frac{\pi \ell t}{2}\right) dt = - \int_{-1}^1 \cos\left(\frac{\pi \ell t}{2}\right) dt \\
 &= -2 \int_0^1 \cos\left(\frac{\pi \ell t}{2}\right) dt \stackrel{\ell \neq 0}{=} -2 \left[\frac{\sin\left(\frac{\pi \ell t}{2}\right)}{\frac{\pi \ell}{2}} \right]_0^1 \\
 &= -\frac{4}{\pi \ell} \sin\left(\frac{\pi \ell}{2}\right) \quad \text{pro } \ell \in \mathbb{N}.
 \end{aligned}$$

$$a_0 = -2 \int_0^1 dt = -2.$$

$$b_\ell = 0 \quad \text{pro } \ell \in \mathbb{N}, \text{ weil } f \text{ je gerade f\u00fcnde.}$$

F. \u00fcr f\u00fcnde f je

$$-1 + \sum_{\ell=1}^{\infty} \frac{-4 \sin\left(\frac{\pi \ell}{2}\right)}{\pi \ell} \cos\left(\frac{\pi \ell t}{2}\right).$$

Sei \u00fcr F. \u00fcr f\u00fcnde f na [-2, 2) je

$$\tilde{f}_f(t) = \begin{cases} -2 & , t \in (-1, 1), \\ -1 & , t \in \{-1, 1\}, \\ 0 & , t \in [-2, -1) \cup (1, 2). \end{cases}$$